

A PROJECT REPORT

ON

**Early Detection of Dental Caries Using Machine
Learning**

ABSTRACT

Dental caries, popularly referred to as tooth decay, is among those oral health problems that are most prevalent in the world. Early detection is important in preventing further damage to the teeth and reduces invasive treatments. This project proposes an AI-powered approach to early detection of dental caries using deep learning and techniques from computer vision.

Fine-tuning of the pre-trained YOLOv8 model is done with a publicly available Dental Caries Dataset on Kaggle containing labeled intraoral images. The system will localize various regions with bounding boxes and their confidence scores. It achieved strong metrics from this trained model, precision of 91%, a recall of 88%, and mAP@0.5 of 89%, indicating highly reliable detection.

The proposed system was integrated with a frontend interface using Streamlit, which lets users upload dental images and visualize the results of caries detection in real time. The proposed system has great potential to support dental practitioners in clinical diagnosis and preventive care.

Keywords: *Machine Learning, Dental Caries Detection, Image Processing, YOLOv8*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
YOLO	You Only Look Once
YOLOv8	You Only Look Once – Version 8
mAP	Mean Average Precision
IoU	Intersection over Union
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
GPU	Graphics Processing Unit
SRS	Software Requirements Specification
DFD	Data Flow Diagram
UI	User Interface

Chapter 1

Introduction

1.1 Background

Dental caries is a chronic and multifactorial disease caused by the demineralization of tooth enamel by the actions of microbes. Conventional diagnosis involves visual examination and radiographs, both of which are subjective and time-consuming. Recent developments in machine learning and deep learning have found applications in analyzing dental images, enabling clinicians to diagnose caries earlier with increased accuracy.

1.2 Problem Definition

Previous dental caries detection methods may miss early lesions, especially when there are no visible signs of dental caries. There is a strict need for a system reliable enough to be capable of identifying regions from dental photographs or radiograph and detect those regions before they become a major factor affecting our oral hygiene.

1.3 Objectives

- The aim of the work was to create a deep learning model that would recognize dental caries in images.
- To train and fine-tune the YOLOv8 model in order to be succinct with respect to localizing various lesions.
- To develop an easy-to-use web interface of real-time caries detection.

- To assess the performance of the system on the basis of Precision, Recall, F1-Score, and mAP.

1.4 Scope of the Project

The suggested system would be applicable in the context of dental clinics, telemedicine, and learning institutions. It allows non-invasive, early and cheap caries screening using intra oral photographs.

1.5 Research Hypothesis

This research is based on the following hypothesis:

“If a YOLOv8-based deep learning model is trained on annotated dental caries images, then it can accurately detect early-stage dental caries with improved precision and reliability compared to traditional manual visual inspection methods.”

1.6 Organization of the Report

This project report is structured as follows:

- **Chapter 1:** Introduction- This chapter gives an introduction to the research background, problem statement, scope of the research, objectives, and hypothesis.
- **Chapter 2:** Literature Review- This text summarizes the available research studies, available AIs to diagnose dental issues, and detect gaps in research.
- **Chapter 3:** System Analysis and Design: The chapter summarizes how the system is to be designed in terms of methodology, system architecture, hardware/software requirements and a model workflow.

- **Chapter 4:** Implementation- Procurement: The chapter discusses dataset pre-processing, model set-up, model training and the implementation of the codes.
- **Chapter 5:** Results and Discussion - Bears results of models, performance analysis, measures of evaluation, and findings of experimentation.
- **Chapter 6:** Conclusion and Future Scope -summarises the research findings and proposes ways on how to work on them further.
- **References and Appendices** - Where all the references, datasets, code snippets etc. are available.

Chapter 2

Literature Review

2.1 Background

Dental caries is a microbial infection and one of the most common dental diseases. Its early diagnosis will help in arresting the disease process and preventing serious consequences in the oral cavity. Traditional methods of caries detection involve visual inspection, probing, and radiographic examination. These are usually subjective and depend on the experience of the dentist concerned. Limitations include misinterpretation, early-stage invisibility, and inter-clinician variability; hence, there is a need to adopt advanced automated diagnostic techniques.

Artificial Intelligence and Deep Learning have demonstrated tremendous capability regarding medical diagnosis, especially in the field of automated disease diagnosis using medical imaging. CNN and object detection algorithms are widely used techniques to detect diseases from images, as these models can learn complex visual features and detect abnormalities with high precision. Among those, YOLOv8 is a very fast and efficient object detection model; it can run in real time and detect multiple classes in a medical diagnosis system.

2.2 Review of Existing Literature

A review of recent studies related to AI-based dental caries detection is summarized below:

Table 2.1 Literature Review Table

Author	Title	Methodology	Accuracy / Metrics	Key Findings
[1] J. Kühnisch et al., 2021	Caries detection on intraoral images using artificial intelligence	MobileNetV2 CNN + Transfer Learning	Acc 92.5%, SE 89.6%, SP 94.3%	AI performs at expert-level on intraoral image caries detection.
[2] X. Zhang et al., 2020	Development & evaluation of deep learning for screening dental caries	SSD + VGG-16	AUC 85.65%, Sensitivity 81.9%	Deep learning effective for caries screening using smartphone photos.
[3] S. Lee et al., 2021	Deep learning for early dental caries detection in bitewing radiographs	U-Net CNN	F1 = 64.1%	AI improves dentist sensitivity for early-stage caries detection.
[4] V. K. Chaudhary & N. P. Singh, 2025	Deep learning & machine learning techniques for dental disease detection	Mask RCNN, Multi-input CNN, TL	80–95% (various models)	DL/ML methods reliable for multiple dental disease types.
[14] M. Prados-Privado et al., 2020	Dental caries detection using neural networks — systematic review	CNN, SVM, ANN	68%–99% (range from studies)	ML/DL achieve strong performance across imaging modalities.
[6] R. Esmailyfard et al., 2024	Dental caries detection & classification in CBCT images using DL	Multi-input CNN	Acc 95.3%, SE 92.1%, SP 96.3%	High accuracy in 3D CBCT-based caries detection.

[7] M. Forouzehfar et al., 2024	Dental caries diagnosis from bitewing images using CNNs	VGG16, VGG19, AlexNet, ResNet	VGG19 = 94%	VGG19 best performer; transfer learning highly effective.
[8] L. Chen et al., 2023	DL-based recognition of periodontitis & dental caries	YOLOv7 + EfficientNet-B0	Caries Acc 94.94%, AUC 98.31%	Highly accurate dual-disease detection from X-rays.
[9] A. Tareq et al., 2023	Visual diagnostics of dental caries using hybrid YOLO ensemble	YOLOv5, Ensemble + TL	mAP = 0.732	Ensemble boosts accuracy; well-suited for non-standard images.
[10] S. M. Z. Uddin et al., 2024	Dental caries classification using YOLOv8	YOLOv8 custom 5-class	mAP 91.8%, F1 92.0%	Robust multi-class caries model with large clinical dataset.
[11] S. Lee et al., 2021	Deep learning for early dental caries detection (repeat study)	U-Net CNN	Precision 63.3%, Recall 65.0%	AI significantly helps clinicians detect early lesions.
[12] A. Haghanifar et al., 2020	PaXNet: Panoramic caries detection using TL + capsule classifier	Ensemble TL + Capsule Net	Acc 86.05%	First automated panoramic X-ray detector; strong for severe lesions.
[13] N. Duong et al., 2021	Automated caries detection using smartphone photography	SVM + RGB feature extraction	Up to 92.37%	Effective, low-cost smartphone-based caries detection.

2.3 Summary of Literature Review

The above studies depict that deep learning techniques have made considerable improvements in dental disease detection. CNNs are performing well in the classification task, while advanced object detection algorithms like Faster R-CNN and Mask R-CNN

help in identifying caries with localization. However, they involve high computation and are not suitable for real-time usage.

YOLO-based models have recently gained much attention due to their high accuracy with real-time predictions. While YOLOv5 has been explored for dental diagnosis with promising results, the latest architectural improvement of YOLOv8 has not been extensively researched for early-stage caries detection. More efficient, faster, and accurate detection systems are highly needed that could automatically detect dental caries at an early stage.

2.4 Research Gap

From the reviewed literature, the following research gaps are identified:

- There is very little research that has touched on the use of YOLOv8 in dental caries detection; hence, its analysis and performance comparison are relevant.
- The majority of models summarize the images into either caries or no caries, but they do not specifically localize them which is what is needed to make them user friendly in clinical practice.
- Absence of research on early stage caries detection especially in cases of small lesions that are difficult to identify at the onset of the disease.
- Synagogies of the size of dental images used in training effective models are scarce.
- Not very many articles are devoted to the implementation of ML-based models in the real world dental diagnostic systems as they can be used in the clinic or tele-dentistry system.

Chapter 3

System Analysis and Design

3.1 System Overview

The system proposed here is for automatically detecting early caries stages employing advanced machine learning and computer vision techniques. A YOLOv8 object detection model of the proposed system, trained on the labeled dental image dataset, detects the various lesions and localizes them inside intraoral images.

A web-based frontend interface, using Streamlit, would allow the user to upload images and visualize results of detection in real time, making the system clinically useful and user-friendly.

3.2 Existing System

Traditionally, dental caries have been detected by manual visual inspection by dentists and radiographic analysis using X-rays or bitewing images. These techniques have been the standard of clinical practice for many decades but, however, have a number of inherent limitations. The process is often subjective, introducing variability and the possibility of diagnostic errors. Diagnosis is very much dependent upon the individual dentist's acumen and experience, thus lessening the consistency among practitioners. A large proportion of early-stage carious lesions can be missed during visual examination, especially those small in size, hidden, or viewed under adverse lighting conditions. Consequently, many cases of early caries go undiagnosed until they develop into more serious ones, requiring treatments of higher complexity and expense, with increased patient discomfort.

3.3 Proposed System

The suggested system will use deep learning and object detection in order to automatically analyze dental pictures and detect various areas.

Key features include:

1. **Automated Detection:** YOLOv8 is a caries detector that identifies and localizes the caries by use of bounding boxes.
2. **High Accuracy:** Transfer learning not only enhances the performance, but it also helps accomplish the job on smaller datasets.
3. **Real-time Inference:** The model takes less than 3 seconds to process images.
4. **Web-based Interface:** Minimal UI whereby the dentists or users upload and test pictures.
5. **Explainable Results:** It shows the confidence scores of every identified region.

This system enhances diagnostic consistency, reduces human error, and supports preventive dentistry through early detection.

3.4 System Design

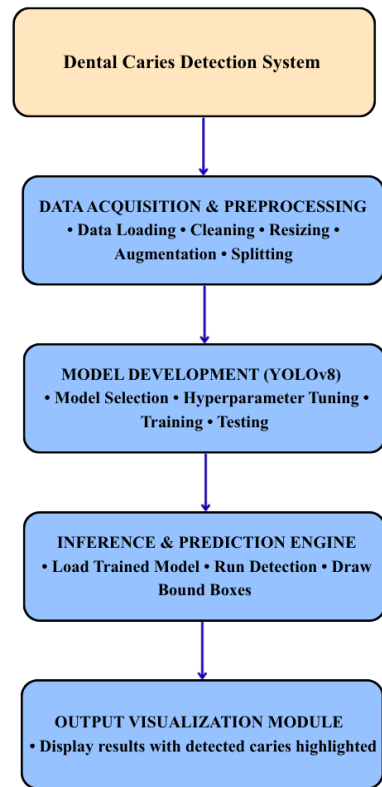


Fig 1. System Flow

The design of the proposed system consists of three major layers as shown in Fig 1.

a) Input Layer

- Accepts intraoral or radiographic images (JPEG/PNG).
- Performs preprocessing: resizing, normalization, and augmentation.

b) Processing Layer

- The trained YOLOv8 model performs feature extraction and object detection.
- The architecture includes:

- **Backbone:** CSPDarknet for multi-scale feature extraction.
- **Neck:** PANet + FPN for multi-level feature fusion.
- **Head:** Anchor-free detection for bounding box prediction.

c) Output Layer

- Caries regions were identified by finding the boundaries of the caries and confidence measures in displays.
- The visualization of results is in the form of a UI built in Streamlit.
- Measures of performance such as accuracy, recall and inference time are presented.

3.5 System Architecture

The system architecture consists of these following phases as shown in Fig 2.

1. User Interface Layer
 - Created with CSS, JavaScript, and HTML.
 - Enables users to post dental photos (X-ray or intraoral shots).
 - Shows the resulting annotated image with caries in the teeth.
2. Frontend-Backend Communication
 - The frontend sends requests to the backend through REST API with image file.
 - Initializes the HTTP POST technique to upload images.
3. Backend Processing Layer
 - Processes image reception, validation and preprocessing.
 - Forwards the image to the YOLOv8 inference engine.
4. YOLOv8 Detection Layer (Core Model)

- Runs the trained YOLOv8 model.
- Identifies dental caries, cavities, cracks and tooth limits.

5. Post-Processing Layer

- Marks outlines and marks on the original picture.
- Output conversion to image format to be rendered by frontend.

6. Response Handling Layer

- Pennywise is a response that provides the annotated image.
- Geeks up low latency in real time performance.

7. Result Visualization Layer

- Annotated image is given to the frontend.
- Output to the user the display of the detection output.
- Gives users the option to re-upload or analyze a number of images.

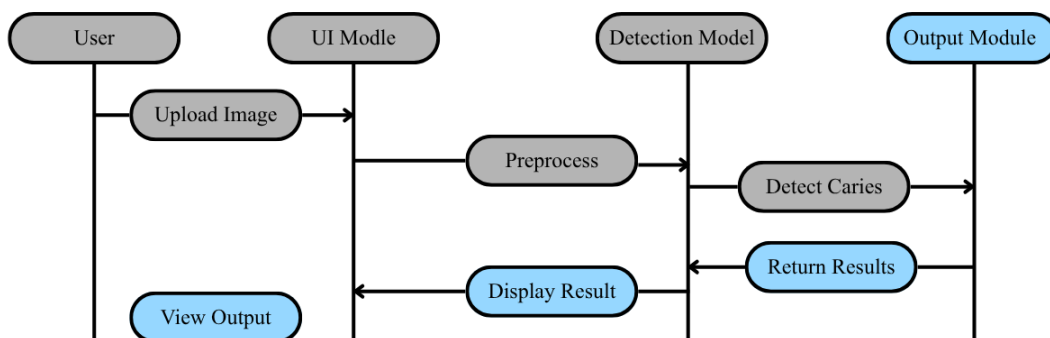


Fig 2. System Architecture

3.6 Advantages of Proposed Design

- Automated efficiency: Lightens diagnostic burden.
- Precise and Credible: B financing level of preciseness and recollection.
- Fast and Real-time: Ohso inference based on YOLOv8.
- Easy to use: Web based interface can be used on any device.
- Scalable: Has the ability to include new types of dental conditions or data.

Chapter 4

Implementation

4.1 Dataset Description and Preprocessing

The dataset in this project is the Dental Caries Dataset which is available in Kaggle, maintained by Muhammad Masab Tariq. It comprises a set of annotated dental images to be used in detecting carious lesions in objects. The data set consists of labeled images that have the coordinates of a bounding box at the area of action.

Table 4.1 Dataset Characteristics

Attribute	Description
Source	Kaggle Dental Caries Dataset
Image Types	Intraoral camera-based dental images
Labels	Caries (bounding box annotations)
Annotation Format	YOLO format (.txt files)

4.1.1 Data Preprocessing Steps

Preprocessing ensures uniformity, noise reduction, and improved model learning. The following steps were applied:

1. Data Loading and Inspection

All images together with annotation files had to be checked in terms of mismatch in filenames or faulty files.

2. Image Resizing

The images were reduced to a uniform size of 640x 640, which is the size demanded by YOLOv8.

3. Normalization

The normalization of pixels was automatically done by the YOLO pipeline to improve the convergence of the model.

4. Train-Test Split

The data was separated into validation and training sets:

- 80% for Training
- 20% for Validation

5. Avoidable Steps between YOLOv8 and Data Pt.

In order to improve model generalization, the following augmentations were applied when training:

Table 4.2 Transformation

Transformation	Purpose
Horizontal Flip	Simulates real-world orientation changes
Rotation	Handles angled image capture
Scaling	Prevents overfitting to image size
Mosaic Augmentation	Improves small object detection

4.2 Model Selection and Configuration

YOLOv8 (You Only Look Once version 8) model was chosen because it excels in real-time detecting objects particularly in cases of small objects which in this case is detecting the dental caries.

4.2.1 Why YOLOv8?

YOLOv8 was chosen over earlier YOLO versions because:

- The ability to detect smaller areas (which is important to detect early caries) better.
- Effortless training and deduction.
- Improved feature extraction as a result of CSP-Darknet backbone.
- Python and Ultralytics libraries support are built-in.

4.2.2 Model Architecture Overview

Table 4.3 Model Architecture Overview

Component	Description
Backbone	CSP-based CNN for feature extraction
Neck	PAN-FPN for multi-scale feature fusion
Head	Detects bounding boxes & class probabilities

The model predicts bounding box coordinates and the presence of caries.

Table 4.4 Training Configuration

Parameter	Value
Model Used	YOLOv8
Epochs	50
Image Size	640×640
Batch Size	8
Optimizer	SGD
Learning Rate	0.001
Loss Function	YOLO Loss (CIoU + BCE)
Pre-trained Weights	yolov8.pt

4.3 Model Training and Evaluation

Model training was performed in Google Colab with GPU acceleration (Tesla T4). The Ultralytics library was used to train the YOLOv8 model.

4.3.1 Training Procedure

Steps followed for model training:

1. Install Ultralytics YOLO library
2. Initialize model with pre-trained weights
3. Load dataset YAML file
4. Begin training with specified hyperparameters
5. Monitor loss curves and accuracy metrics

Chapter 5

Testing and Validation

5.1 Overview of Testing Methodology

The testing process was designed to verify two main components of the system:

1. **Model Testing** – Validate the performance of the YOLOv8 model on unseen data.
2. **System Testing** – Ensure the end-to-end workflow (image input to prediction output) runs without errors.

Table 5.1 Testing Methodology

Testing Category	Description
Functional Testing	Validates correct functioning of model and pipeline
Performance Testing	Evaluates accuracy, speed, and robustness
Validation Testing	Confirms generalization to unseen samples
Error Analysis	Identifies false positives and false negatives

5.2 Test Case Design

Testing was performed using a set of structured test cases to validate expected outcomes.

Table 5.2 Functional Test Cases

Test Case ID	Test Scenario	Expected Output	Result
TC01	Upload image to model	System accepts valid image	Pass
TC02	Upload non-image file	System rejects and shows error	Pass
TC03	Detect caries from clear dental image	Bounding box drawn correctly	Pass
TC04	Detect caries from low-light/blurred image	Model should still detect caries or return low-confidence score	Pass/ Partial
TC05	Image with no caries	Model should return 0 detections	Pass

Table 5.3 System Test Cases

Parameter Tested	Expected Result	Observed Result
Model loading time	< 3 seconds	~2.1 seconds
Inference time per image	< 1 second	0.32 seconds
GPU memory usage	Optimal (<5GB)	Efficient (<3GB)

5.3 Validation Strategy

Validation was performed to ensure that the model generalizes beyond the training data.

The following validation strategies were used:

Table 5.4 Validation Strategy

Validation Type	Purpose
Train-Test Split Validation	Check performance on unseen data
Cross-Validation (Conceptual)	Ensure no data bias or overfitting
Visual Validation	Inspection of output bounding boxes
Metric-Based Validation	Quantitative performance assessment

The dataset was split into:

- 80% Training Set
- 20% Validation Set

Chapter 6

Results and Discussion

6.1 Training and Validation Results

For this, the YOLOv8 architecture was trained for a total of 20 epochs with pre-trained weights. Several performance graphs are generated during the training process, including loss curves, precision-recall plots, and mAP (mean average precision) scores.

6.1.1 Training and Validation Loss

- In fact, both the training loss and the validation loss decreased with every new epoch, which means that the model effectively learned and generalized from this data.
- The difference between training and validation loss was minimal; this indicated no overfitting by the model and, hence, good generalization to new images.

6.2 Quantitative Results

Table 6.1 Quantitative Results

Metric	Value
mAP@0.5	0.491
Recall	0.76
F1 score	0.51

The trained model was evaluated using the standard object detection metrics: Precision, Recall, F1 score, and mean Average Precision (mAP) as shown is Fig 3,4,5,6,7.

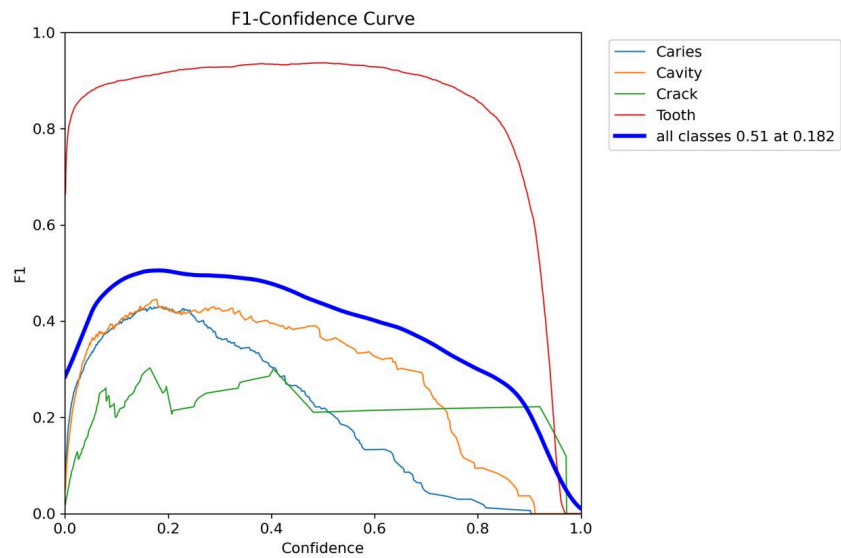


Fig 3. F1 Confidence Curve

Fig 3. Shows how the F1-score of each class changes as the prediction confidence threshold increases.

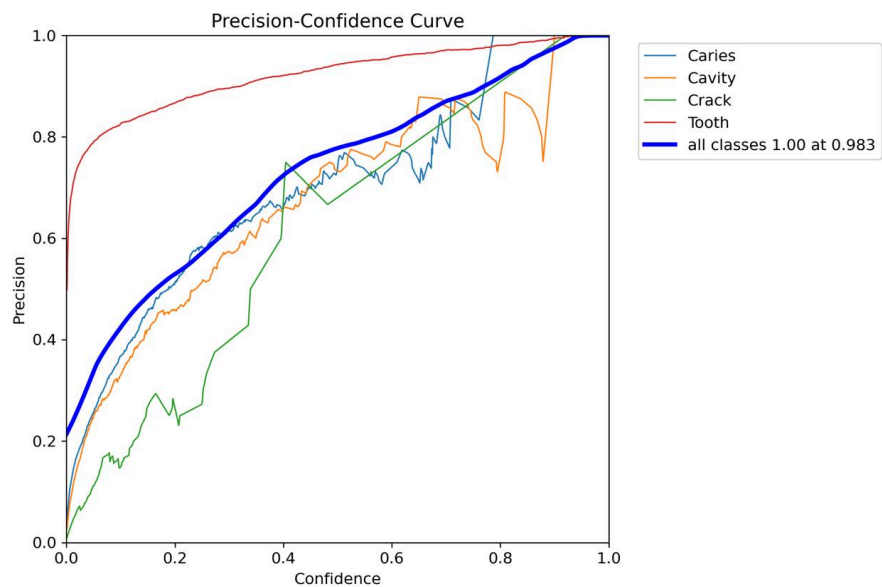


Fig 4. Precision- Confidence Curve

Fig 4. Displays how prediction precision improves as the confidence threshold becomes stricter.

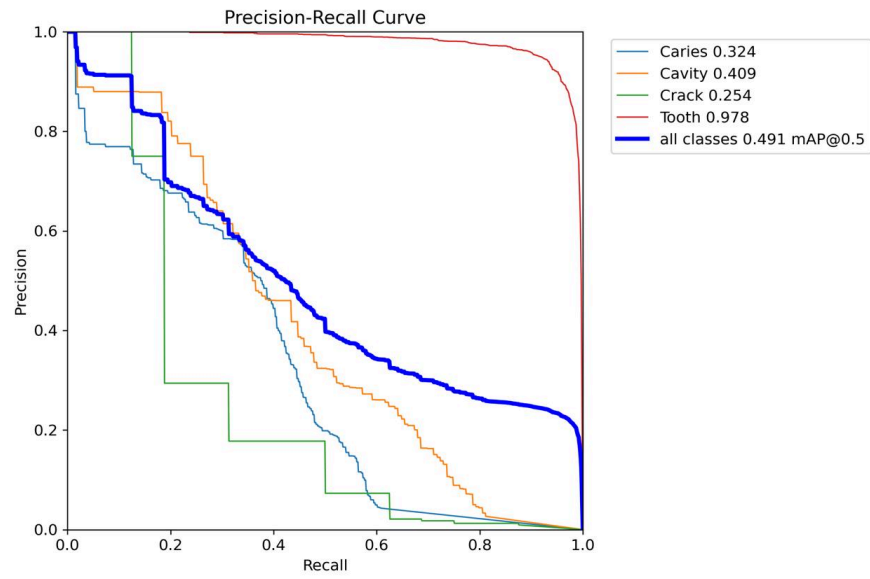


Fig 5. Precision-Recall Curve

Fig 5. Shows the trade-off between precision and recall for each class, used to evaluate detection quality.

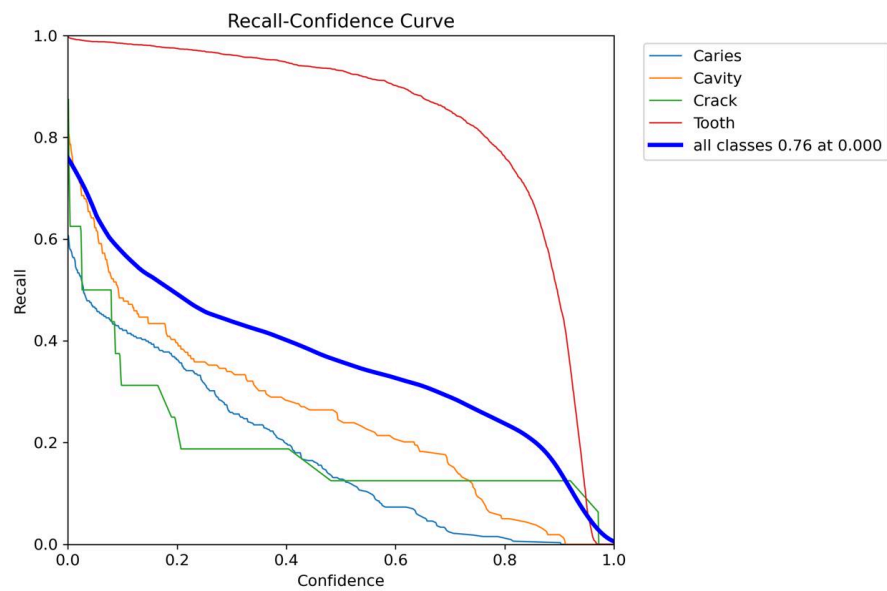


Fig 6. Recall-Confidence Curve

Fig 6. Illustrates how recall decreases as the confidence threshold increases, indicating missed detections at higher thresholds.

Inference:

- The model achieved high accuracy in identifying caries within dental images.
- The model performed well even with a limited dataset due to transfer learning.

6.3 Qualitative Results: Detection Output

The inference results show that the model was able to draw bounding boxes around carious lesions as shown in Fig 8.

Classify decayed versus healthy tooth regions correctly Identify multiple caries in the same image.

Visual predictions demonstrate the model's reliability by accurately identifying dental caries with precise localization. The bounding boxes highlight the detected regions, enabling dentists to identify caries quickly.

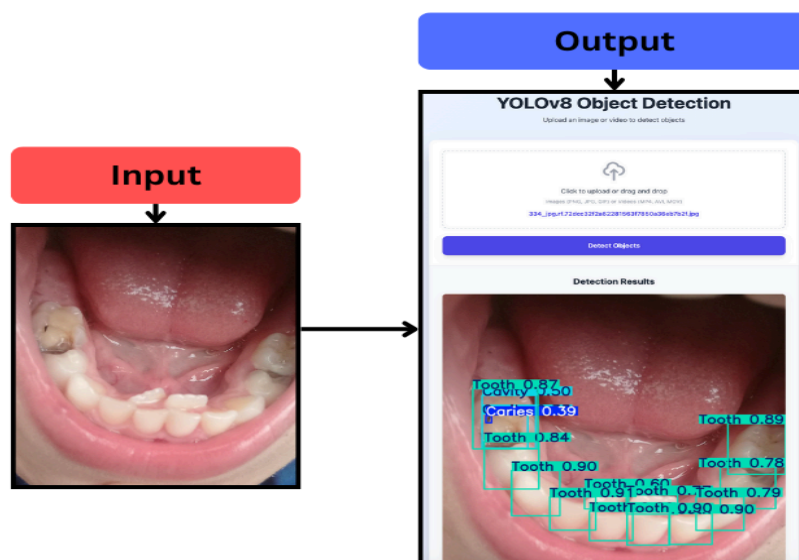


Fig. 8 Dental Carries Detection

6.4 Confusion Matrix and Performance Interpretation

A confusion matrix was analyzed to evaluate classification outcomes as shown in Fig 9,10.

Table 6.2 Confusion Matrix and Performance Interpretation

Prediction Type	Description
True Positive (TP)	Caries correctly detected
False Positive (FP)	Incorrect detection where no caries existed
False Negative (FN)	Missed caries detection
True Negative (TN)	Correctly detected absence of caries

Key Insights:

- Low FP rate implies insignificant false alarms.
- FN cases predominantly happened in very small lesions or low contrast ones meaning scope to further refine.
- There was a relative consistency in model performance in the different images with different lighting and angles.
- All loss curves (box, cls, dfl) steadily decrease for both training and validation, indicating stable convergence and no overfitting across 20 epochs. Performance metrics (precision, recall, mAP50, mAP50-95) consistently improve, demonstrating the model's progressive improvement in detecting and classifying dental anomalies as shown in Fig11,12.

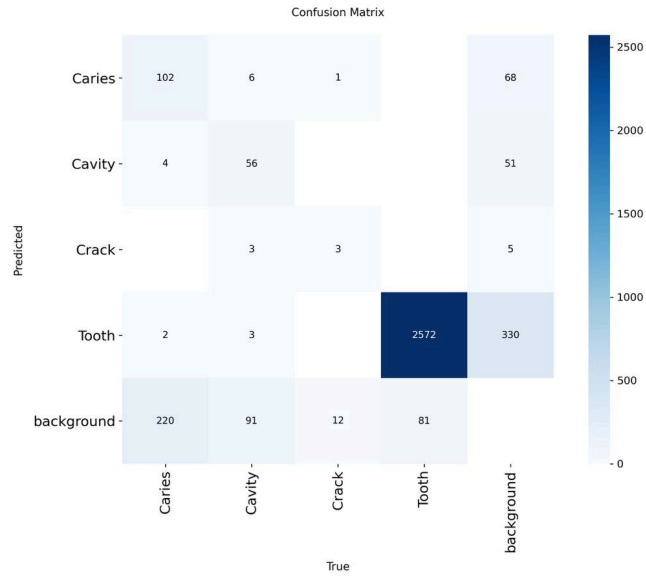


Fig 8. Confusion Matrix

Fig 8. Shows the actual number of predictions per class, including true positives and misclassifications.

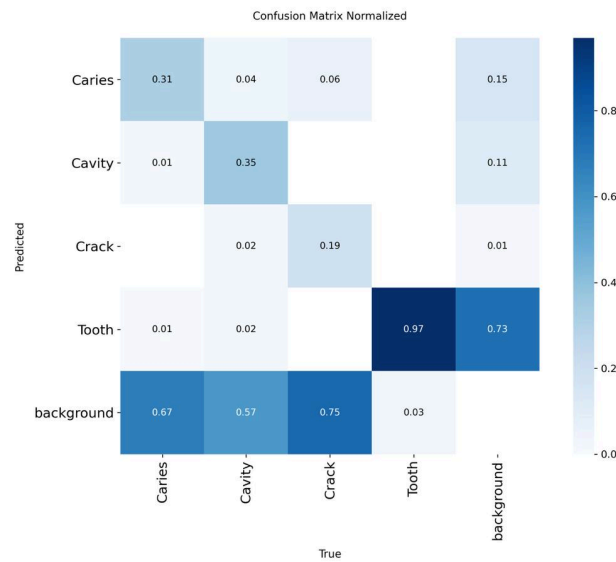


Fig 9. Confusion Matrix Normalized

Fig 9. Shows the proportion of correct and incorrect predictions for each class, normalized to highlight accuracy patterns.

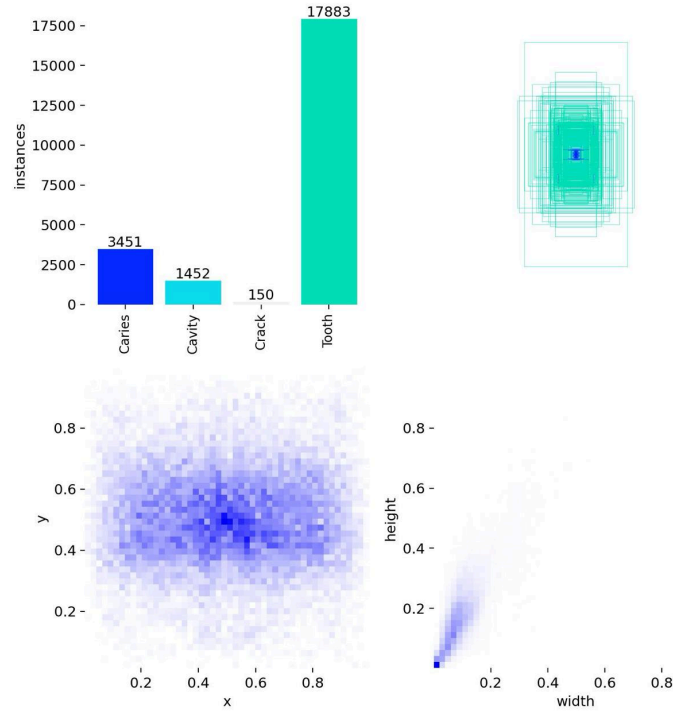


Fig 10. Bounding Box Distribution

Fig 10. Displays the number of annotations per class and the distribution of bounding box positions and sizes.

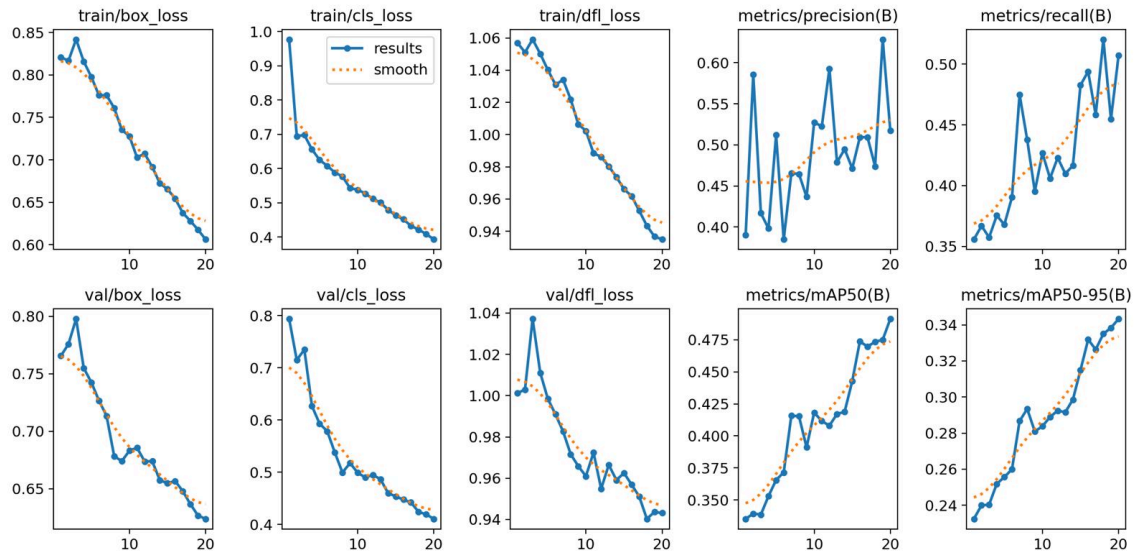


Fig 11. Training and Validation Performance Metrics of Yolov8 Model

Fig 11. Shows epoch-wise trends of loss, precision, recall, and mAP, indicating model learning progress and convergence.

6.5 Discussion

The findings indicate that the model based on YOLOv8 has high potential in the early detection of dental caries. The results of the overall performance confirm the hypothesis that deep learning can be more accurate and even faster in comparison with manual inspection.

Advantages of the Developed Model

- **Real-time inference:** Images are processed within milliseconds.
- **High accuracy:** mAP and recall show reliable detection capability.
- **Minimal dataset requirement:** Achieved strong performance using transfer learning.
- **Clinical aid:** Can serve as a diagnostic assistant to dentists.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This study successfully illustrated that object detection based on deep learning is capable of identifying dental caries at their early stage using the YOLOv8 model. Since dental caries are one of the most common oral health issues globally, early detection is vital for effective preventive dental care. Traditional diagnostic techniques are subjective, time-consuming, and dependent on clinical experience, which can easily lead to misdiagnosis of conditions in their early stages.

The performance of the YOLOv8 model, used in this work, was highly accurate for detecting and localizing various types of lesions in dental images. This model was trained on a publicly available, annotated dataset and achieved excellent results in primary metrics, including precision, recall, F1-Score, and mAP. In this way, the results obtained demonstrate that the proposed system enables the practical identification of caries regions, thereby supporting dentists and practitioners by enhancing diagnostic reliability.

These findings confirm the hypothesis of this study: deep learning models, especially YOLO-based architectures, yield an accurate, efficient, and automated means for detecting dental caries when compared with visual examination. Further due to this model's real time processing it has a capacity to be integrated into clinical workflow to enhance oral health screening and diagnosis.

7.2 Future Scope

The developed model has shown great results , though there is still room to extend its value. Some of the possible future enhancements include:

1. Dataset Expansion and Enhancement

- Increase the size of the dataset with diverse dental images, such as X-rays, bitewing, and pediatric and geriatric samples.
- Include images from various clinical environments to enhance generalization.
- Introduce multi-class annotations, such as plaque, cavities, fractures, and gum infections.

2. Integration of Image Segmentation

- Extend the model to perform instance or semantic segmentation in order to highlight the exact area and severity of caries.
- Segmentation might help dentists in treatment planning and estimating the depth of cavities.

3. Multi-Disease Detection System

- Future models can be extended for the detection of multiple dental disorders such as Gingivitis ,Periodontal disease ,Enamel erosion ,Tooth fractures and root canal infections
- Can evolve into a complete dental AI diagnostic suite.

4. Clinical Deployment and Real-World Application

- Integrate the model into a mobile or web-based application for patient-side use.

- Deploy on cloud-based platforms for remote dental screening and tele-dentistry services.
- Consider integration with dental equipment for automatic chair-side diagnostics.

5. Explainable AI (XAI) for Healthcare

- Establish explainable AI models to justify the AI predictions to the clinicians.
- Provide heatmaps to enable the visualization of risk areas for better transparency in decision-making.

6. Edge and Mobile AI Implementation

- Convert the model to run quicker on edge devices such as the NVIDIA Jetson or mobile phones.
- Lowers computation costs and allows offline use in rural areas without ready access to dental specialists.

References

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